

Characterization of Root Exudates at Different Growth Stages of Ten Rice (*Oryza sativa* L.) Cultivars

M. S. Aulakh^{1,2}, R. Wassmann^{1,3}, C. Bueno¹, J. Kreuzwieser⁴, and H. Rennenberg^{3,4}

¹ International Rice Research Institute, Soil and Water Sciences Division, Makati City, Philippines

² Present address: Department of Soils, Punjab Agricultural University, Ludhiana, Punjab, India

³ Fraunhofer Institute for Atmospheric Environmental Research, Garmisch-Partenkirchen, Germany

⁴ Institute for Forest Botany and Tree Physiology, University of Freiburg, Freiburg, Germany

Received: January 8, 2001; Accepted: February 13, 2001

Abstract: Plant root exudates play important roles in the rhizosphere. We tested three media (nutrient solution, deionized water and CaSO₄ solution) for three periods of time (2, 4 and 6 h) for collecting root exudates of soil-grown rice plants. Nutrient culture solution created complications in the analyses of exudates for total organic C (TOC) by the wet digestion method and of organic acids by HPLC due to the interference by its components. Deionized water excluded such interference in analytical analyses but affected the turgor of root cells; roots of four widely different rice cultivars excreted 20 to 60% more TOC in deionized water than in 0.01 M CaSO₄. Furthermore, the proportion of carbohydrates in TOC was also enhanced. Calcium sulfate solution maintained the osmotic environment for root cells and did not interfere in analytical procedures. Collection for 2 h avoided under-estimation of TOC and its components exuded by rice roots, which occurred during prolonged exposure. By placing plants in 0.01 M CaSO₄ for 2 h, root exudates of soil-grown traditional, tall rice cultivars (Dular, B40 and Intan), high-yielding dwarf cultivars (IR72, IR52, IR64 and PSBRc 20), new plant type cultivars (IR65598 and IR65600) and a hybrid (Magat) were collected at seedling, panicle initiation, flowering and maturity and characterized for TOC and organic acids. The exudation rates were, in general, lowest at seedling stage, increased until flowering but decreased at maturity. Among organic acids, malic acid showed the highest concentration followed by tartaric, succinic, citric and lactic acids. With advancing plant growth, exudation of organic acids substituted exudation of sugars. Root and shoot biomass were positively correlated with carbon exudation suggesting that it is driven by plant biomass. As root exudates provide substrates for methanogenesis in rice fields, large variations in root exudation by cultivars and at different growth stages could greatly influence CH₄ emissions. Therefore, the use of high-yielding cultivars with lowest root excretions, for example IR65598 and IR65600, would mediate low exudate-induced CH₄ production. The screening of exciting rice cultivars and breeding of new cultivars with low exudation rates could offer an important option for mitigation of CH₄ emission from rice agriculture to the atmosphere.

Key words: Root exudates, organic carbon, organic acids, carbohydrates, rice cultivars, growth stages, methane production.

Introduction

On average, 30–60% of photosynthesized C is allocated by plants to the roots and a substantial portion of this C is released or secreted by the roots in the form of organic compounds in the rhizosphere (Marschner, 1995^[31]). Root exudates play a decisive role in solubilizing and mobilizing nutrients in the soil, thereby enhancing acquisition of nutrients by plants, in immobilizing toxic elements for protecting plants against toxicity (Jauregui and Reisenauer, 1982^[19]; Hoffland et al., 1992^[17]; Jones and Darrah, 1994^[22]; Beissner and Römer, 1998^[8]; Kirk et al., 1999^[23]) and in providing substrates for microbial activity in the rhizosphere (Andal et al., 1956^[1]; Mac Rae and Castro, 1966^[30]). In flooded rice soils, root exudates provide important C sources for methane (CH₄) production (Dannenberg and Conrad, 1999^[11]; Lu et al., 2000^[29]). Organic acids in root exudates supply energy to soil microbial communities, including methanogens, and also mobilize soil phosphorus (Beissner and Römer, 1998^[8]; Kirk et al., 1999^[23]) and micronutrients (Marschner et al., 1989^[32]; Zhang et al., 1989^[45]). Several greenhouse and field measurements showed a large CH₄ flux at the late development stage of rice plants, which has been attributed to the supply of organic C from rice root exudation (Holzapfel-Pschorn et al., 1986^[18]; Lindau et al., 1991^[27]; Rennenberg et al., 1992^[37]; Wang et al., 1997^[42]). The disparity in CH₄ production levels between vegetated and un-vegetated soils has been ascribed to the contribution of root exudates and root autolysis products towards the substrates available for methanogenesis (Kludze et al., 1993^[24]).

The collection and characterization of root exudates have been extensively studied in upland plants and trees using several methods of plant cultivation (i.e., hydroponic culture, soil-grown), collection media (i.e., nutrient culture solution, CaCl₂, deionized water) and collection periods varying from 2 to 6 h (Jones and Darrah, 1992^[20], 1993^[21]; Vaughan et al., 1994^[41]; Marschner, 1995^[31]; Dinkelaker et al., 1996^[12]; Hodge et al., 1996^[16]; Beissner and Römer, 1998^[8]; Neumann and Römhild, 1998^[34]). On the other hand, very few reports have directly collected and examined root exudates of rice plants that are grown under flooded conditions (see review by Wassmann and Aulakh, 2000^[43]).

The analytical methods used for detecting exudates may be impeded by ions present in the nutrient culture solution. Lu et al. (1999^[28], 2000^[29]) used deionized water for collecting

exudates. On the other hand, deionized water increases the turgor of root cells, and thus, may eventually affect exudation. Exudation collection times varying from 2 to 6 h have previously been used to collect plant root exudates (Hodge et al., 1996^[16]; Beissner and Römer, 1998^[8]; Neumann and Römheld, 1998^[34]; Lu et al., 1999^[28], 2000^[29]). As the majority of organic C compounds present in exudates are easily decomposable, prolonged collection times may result in the loss of easily degradable C, leading to an under-estimation of C released by plant roots. Therefore, there is a need to find appropriate exudate collection media and exposure times in order to develop an efficient method for collection, quantification and characterization of root exudates of plants.

Considering the enormous genotypic and phenotypic variations in the species *Oryza sativa* (Leon and Carpena, 1995^[25]), large variations in the quantity (exudation rate) and quality (composition) of exudates have to be taken into account. However, such information is not available in the literature. A thorough understanding of the quality and quantity of root exudates of rice during different growth stages and of widely-used cultivars could help in selecting and breeding cultivars that minimize root exudation and, as a consequence, reduce CH₄ emission from rice fields. Since root exudation represents a possible loss of photosynthate from the plant, minimizing this process may also result in increased rice yields. On the other hand, reducing root exudation may have negative effects for nutrient uptake and may cause toxicities.

The objectives of this study were (a) to develop a feasible method for collecting root exudates, (b) to identify the exudation rate and characterize the exudates of 10 rice cultivars during different growth stages, and (c) to identify relationships between root exudation and growth characteristics of plants as a basis for a selection of exciting cultivars and future breeding of cultivars.

Materials and Methods

Rice cultivation

The method for collecting root exudates was developed using the rice (*Oryza sativa* L.) cultivars Dular and Intan (traditional tall cultivars), IR72 (high-yielding dwarf cultivar from the International Rice Research Institute [IRRI] at Los Banos, Philippines) and IR65598 and IR65600 (new plant type [NPT] cultivars). In a subsequent set of experiments, 10 cultivars were selected from four categories: (a) traditional tall cultivars (Dular, B40 and Intan), (b) high-yielding dwarf cultivars from IRRI (IR72, IR52, IR64 and PSBRc 20), (c) NPT cultivars (IR65598 and IR65600), and (d) a hybrid cultivar from IRRI (Magat) for quantification and characterization of exudates. In a greenhouse at IRRI, the seeds of these cultivars were cultivated following the procedure described by Aulakh et al. (2000a^[2]). Briefly, the rice seeds were sown on a framed nylon mesh that was floated on a 3-cm deep nutrient culture solution in a plastic tray. The nutrient solution contained 40 mg N l⁻¹ (as NH₄NO₃), 10 mg P l⁻¹ (as NaH₂PO₄ · H₂O), 40 mg K l⁻¹ (as KCl), 40 mg Ca l⁻¹ (as CaCl₂), 40 mg Mg l⁻¹ (as MgCl₂), 0.5 mg Mn l⁻¹ (as MnCl₂ · 4H₂O), 0.05 mg Mo l⁻¹ (as [NH₄]₆MoO₂₄ · 4H₂O), 0.2 mg B l⁻¹ (as H₃BO₃), 0.01 mg Zn l⁻¹ (as ZnSO₄ · 7H₂O), 0.01 mg Cu l⁻¹ (as CuSO₄ · 5H₂O) and 0.01 mg Fe l⁻¹ (as FeSO₄ · 7H₂O + EDTA in 1 N KOH). Two weeks after seeding, the plants

were transplanted individually into plastic pots (16 cm i.d. × 15 cm length), each lined with a polyethylene bag and filled with 2.3 kg crushed Maahas clay soil collected from a rice field at IRRI. Characteristics of the Maahas clay were determined by standard methods (Westerman, 1990^[44]). The air-dried Maahas clay had a pH of 6.4, CEC of 34.1 cmole kg⁻¹, available P of 19 mg kg⁻¹, organic C of 15.7 g kg⁻¹ and total N content of 1.9 g kg⁻¹. Each pot was given 150 kg N ha⁻¹ as urea in three splits, viz. basal, at maximum tillering and flowering stage. Plants were irrigated with water maintaining a 5-cm water layer overlying the soil surface.

Exudate collection, processing and analysis

In the experiments, the rice plants were taken out of the pots together with the intact soil and the plastic bag. The plastic bags were cut open and the soil around the roots was washed off with a gentle water spray; final washing was given with distilled water. As every care was taken, this procedure facilitated the removal of soil without injuring the roots. If, inevitably, minor damage was caused to fine roots including possible Actinomyces associations, it should not have seriously modified/biased results. Using a PVC tube, cylindrical pots (6 cm i.d. × 15 cm effective height) were specially designed for collection of root exudates from individual plants. An individual plant was placed in a PVC pot (containing 500 ml of collection media) in such a position that the complete root system of the plant was submerged in the collection media (described below) and the shoots were just protruding through a hole in the centre of the pot lid. After 2, 4 or 6 h, the effluent containing the root exudates was collected from each pot.

Immediately after collection, each sample of exudate solution was successively filtered through a Whatman No. 42 filter paper and a 0.45 µm membrane filter to remove root detritus and microbial cell debris. The filtrate was divided into three portions. One portion was analysed for TOC and carbohydrates. The second portion (200 ml) of each exudate solution was immediately frozen and subsequently concentrated to powder form by freeze-drying. The samples were kept frozen until analysis of organic acids. The third portion of each exudate sample was immediately frozen, concentrated to a volume of 20 ml by freeze drying and used in the experiments on exudate-induced CH₄ production potentials (Aulakh et al., 2001^[5]).

TOC was determined by the wet digestion method, as described in the protocol of Nelson and Sommers (1996^[33]) with few modifications. An aliquot of 5 ml was mixed with 5 ml of 0.035 N K₂Cr₂O₇, 10 ml of 98% H₂SO₄ and 5 ml of 88% H₃PO₄, and kept in a digestion tube at 150 °C for 30 min. The samples were cooled, transferred to 150 ml Erlenmeyer flasks and titrated with 0.005 N Fe(NH₄)₂(SO₄)₂ · 6H₂O in 0.4 M H₂SO₄. Sucrose was used as the reference C substance.

The amount of carbohydrates was determined using the anthrone colourimetric assay (Brink et al., 1960^[9]). An aliquot of 2 ml was mixed with 4 ml anthrone reagent, each tube vortex mixed for 60 sec, cooled to room temperature, and the absorbance was measured at 625 nm using a UV-VIS spectrophotometer (Spectronic 601, Milton Roy Co, USA). The C concentration in the exudate samples was determined by extrapolation of the measured absorbance using a standard curve obtained from the absorbance of glucose (D+) standards.

The exudation rates of TOC and carbohydrate C exuded by rice roots were calculated as follows:

C exudation per plant ($\text{mg C plant}^{-1} \text{d}^{-1}$) =

$$\frac{(\text{C in a sample with a plant } [\text{mg C ml}^{-1}] - \text{C in a control without a plant } [\text{mg C ml}^{-1}]) \times 500 (\text{ml plant}^{-1}) \times 24 (\text{h d}^{-1})}{\text{Collection period (h)}}$$

C exudation per biomass unit ($\text{mg C g}^{-1} \text{ plant biomass d}^{-1}$) =

$$\frac{\text{C exudation per plant } (\text{mg C plant}^{-1} \text{d}^{-1})}{\text{Plant (root or root + shoot) biomass } (\text{g plant}^{-1})}$$

Organic acids were analysed according to a modified protocol of Badoud and Pratz (1986^[6]). Each freeze-dried sample was re-dissolved in 1 to 7 ml distilled water. Aliquots of 20 μl of the solutions were added to 100 μl^{-1} of 1,4-dioxan (Merck, Germany) containing 2 mg O-(4-nitrobenzyl)-N,N'-diisopropylisourea (PNBDI, Sigma-Aldrich, Germany) as a derivatization reagent. After derivatization at 80 °C for 60 min, the samples were cooled to room temperature and 400 μl H₂O and 600 μl acetonitrile (Merck, Germany) were added. Subsequently, 100 mg Dowex 50 W-X8, 100–200 mesh, Sigma, Germany) were added, the samples were shaken for 2 min, incubated for 15 min at room temperature and centrifuged at 14000 rpm for 10 min at 10 °C. Thereafter, 100 μl of the supernatant of each sample were analysed by HPLC (Model-System Gold, Beckman, Germany) using a water/acetonitrile gradient (0–20 min, 20–80% acetonitrile) at a flow rate of 1 ml min^{-1} for elution. Separation of organic acids was performed on a reversed-phase octadecyl-silicium column, C-18, 5 μm , 250 $\times$ 4.6 mm (Ultrasphere™, Beckman, Germany). Organic acids were detected at a wavelength of 265 nm with a diode array detector (Beckman, Germany). Individual organic acids were identified and calibrated by comparing retention times with those of standards prepared with the known amounts of lactic, formic, acetic, tartaric, malic, oxalic, succinic and citric acids which were eluted with a retention time of 11.2, 13.5, 14.4, 15.5, 16.8, 18.8, 19.4 and 20.2 min, respectively. Neither oxalic nor formic acids were detected in any of the exudates. The subtraction of blank values from exudates samples accounted for the organic acids, if any, released by the filters used.

Influence of collection media and period

A first set of experiments included a control (without plant) and plants of the rice cultivar IR72 at two growth stages (seedling and maximum tillering), three collection media (nutrient culture solution, deionized water and CaSO₄ solution) and three collection periods (2, 4 and 6 h) in three replications each. The molar concentration of calcium sulfate (0.01 M) chosen was close to the molarity of the nutrient culture solution, and showed no effect on the turgor of rice root cells by microscopical investigation (not shown). The nutrient solution (described above) and calcium sulfate (0.01 M CaSO₄ · 2H₂O) were prepared in deionized water immediately before use. Preliminary experiments indicated that the composition of root exudates collected in these media was independent of purging with N₂ (used to mimic anaerobic flooded rice soil conditions). Therefore, purging with N₂ was omitted. Exudates were collected and characterized for TOC, carbohydrates and organic acids, as described above, at seedling (plant age 31 d) and max-

imum tillering (plant age 62 d). Thereafter, each plant was divided into roots and shoots (aboveground), and dried in an oven at 110 °C for biomass determination.

The experiments with IR72 plants were repeated in a second set with plants of Dular, Intan, IR65598 and IR65600 at panicle initiation (PI), and two collection media (CaSO₄ solution and deionized water) in three replications each. Exudates were collected for 2 h and characterized for TOC following the protocol described above.

Quantification and characterization of root exudates of 10 cultivars at different growth stages

Root exudates were collected from three plants of each of the above-listed 10 cultivars at four physiological growth stages, viz. seedling stage - when plants initiated tillering, PI, flowering and maturity, following the protocol given above. Every time a control (0.01 M CaSO₄ solution without plant) was included in triplicate. At each sampling, the exudates were collected at the same time of the day (10.00 h to 12.00 h) to avoid any variation by diurnal changes. After collecting root exudates, each plant was divided into roots, shoots and grains (at maturity), and dried in an oven at 110 °C for biomass determination.

Statistical analysis

The data presented are means $\pm$ standard deviation (SD) of three plants. Statistical analysis of experimental data was accomplished by Analysis of Variance in Completely Randomized Design (Cochran and Cox, 1950^[10]) using IRRISTAT statistical software (Bartolome et al., 1999^[7]). Mean separation for different treatments in each experiment was performed using the least significant difference (LSD) test for significance at the 0.05 level of probability. Correlation coefficients (*r*) were determined to show the degree of association between root exudation and different plant parameters.

Results and Discussion

Influence of exudate collection media and time

Analysis of root exudates for organic C by the wet digestion method revealed complications from the composition of the nutrient solution, possibly due to interference of chloride or other ions present (Nelson and Sommers, 1996^[33]). TOC in control (without plant) samples was 30 $\mu\text{g ml}^{-1}$ of nutrient solution as compared to 0.8 $\mu\text{g ml}^{-1}$ in deionized water. Apparently, nutrient solution is not a feasible medium for investigating root-exuded C when TOC is determined by the wet digestion method. The wet digestion method, used to determine organic C in root exudates, was not suitable for analysis of nutrient solution.

Another problem encountered was that nutrient solution gave large background peaks during organic acid analysis by HPLC which overlapped and masked the peaks for several organic acids. As observed by Kirk et al. (1999^[23]), the detection of early-eluting organic acids was not possible due to the presence of early-eluting inorganic ions such as NO₃⁻ in the nutrient solution. We note that analysis with little or no matrix interferences from ions in nutrient/soil solutions may be possible

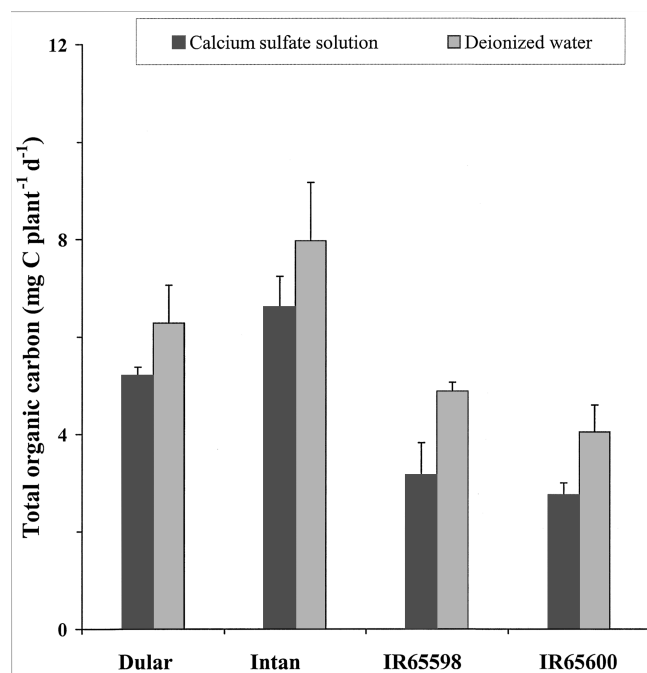


Fig. 1 Comparative root exudation (total organic C) of the rice plants of Dular and Intan (traditional tall cultivars) and IR65598 and IR 65600 (new plant type dwarf cultivars) at panicle initiation stage in deionized water or calcium sulfate solution. Data shown are means $\pm$ SD of 3 replicate plants.

using other techniques, such as C and N analyser and membrane enrichment (Shen et al. 1996^[39]). However, the wet digestion method is an inexpensive, simple technique that is most widely used in plant nutrition laboratories. Therefore, deionized water and CaSO_4 solution were used instead.

Deionized water excluded interference by oxidizable ions but altered the turgor of the root cells. Collection of root exudates of plants from Dular, Intan, IR65598 and IR65600 cultivars, representing two widely different genotypes (tall traditional and dwarf NPT cultivars), at PI stage, showed 20 to 60% more C

released in deionized water than in CaSO_4 solution (Fig. 1, Table 1). We note that when plants are immersed in deionized water, the water will not remain deionized for long due to the excretions by roots. However, this does not preclude initial damage or alteration of exposed root cells resulting in release of more C in deionized water, presumably to equilibrate osmotic pressure or turgor pressure of the cell sap (Giaquinta, 1980^[14]).

The collection media also greatly affected the composition of root exudates (Fig. 2, Tables 1 and 2). In seedlings, while plants released 83% of TOC as sugars in deionized water, only 67% of TOC were released in CaSO_4 solution (Fig. 2). Similarly, at maximum tillering, significantly more sugars were released in deionized water (62%) than in CaSO_4 solution (44%). This shift towards the enhanced release of sugars into deionized water further supported the assumption that deionized water surrounding the roots affected the turgor of root cells. Sugars are highly water-soluble, easily synthesized and efficiently translocated (Fukumorita and Chino, 1982^[13]); plants may increase phloem loading and translocation of sugars in response to decreased osmotic pressure (Giaquinta, 1980^[14]). Exudation of organic acids, however, was lower in deionized water (Tables 1 and 2) for reasons unknown.

Analysis of root exudates for 2, 4 and 6 h in both CaSO_4 and deionized water showed that the longer the period of exposure the lower was the rate of TOC exuded (Fig. 2). While the increase in collection time from 2 h to 4 h resulted in 25 to 34% lower TOC exudation, a further decrease (5–10%) at 6 h was relatively lower. The amounts of carbohydrate C and organic acid C were also reduced with prolonged collection period (Fig. 2). This reduction in TOC, as well as its individual components, in root exudates could be due to the decomposition of easily degradable C compounds by microorganisms present on rice root surfaces. Neumann (pers. communication) observed rapid degradation of organic acids in root exudates of white lupin after an incubation period of 5 and 7.5 h. Rattray et al. (1995^[36]) showed that the incorporation of ^{14}C -labelled exuded material into microbial biomass was greatest after 3 h period, and declined thereafter. Kirk et al. (1999^[23]) investigated the kinetics of citrate decomposition in the soil suspen-

Table 1 Results of analysis of variance of method comparison; different letters indicate significant differences (0.05 level of probability) using calcium sulfate (CS) and deionized water (DW) as medium for the respective combination of carbon fraction, cultivar and sampling time. Different letters indicate significant differences at $p \geq 0.05$

	Total organic carbon		IR65598 CS; DW	IR65600 CS; DW	IR72 CS; DW	Org. acids	Carbon
	Dular CS; DW	Intan CS; DW				IR72 CS; DW	IR72 CS; DW
Panicle initiation							
2 h	a; b	a; b	a; b	a; b			
Seedling stage*							
2 h					a; a	a; a	a; a
4 h					a; a	a; a	a; b
6 h					a; a	a; a	a; b
Maximum tillering*							
2 h					a; a	b; a	a; a
4 h					a; b	b; a	a; b
6 h					a; a	b; a	a; b

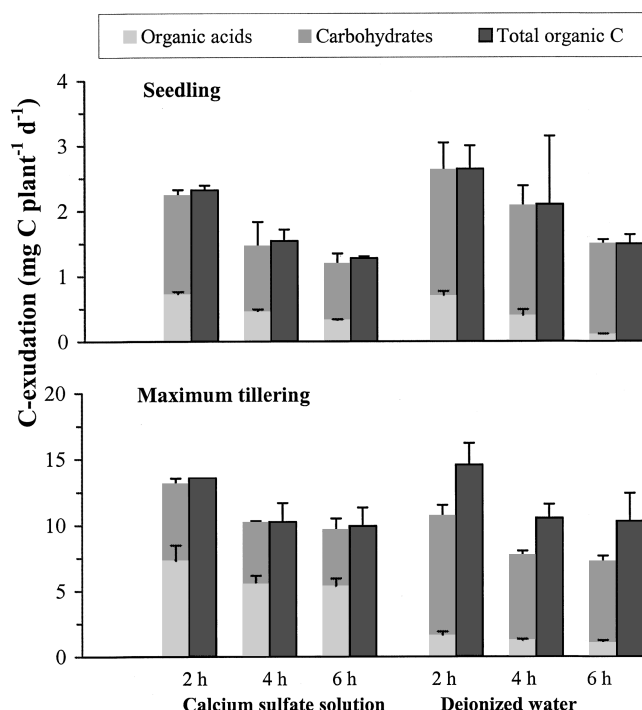


Fig. 2 Total organic C, carbohydrates and total organic acids in root exudates of rice cultivar IR72 at seedling and maximum tillering stages, collected in calcium sulfate solution or deionized water. Data shown are means $\pm$ SD of 3 replicate plants. Different letters on the top of right-side columns (total organic C) indicate significant differences between collection period of 2, 4 and 6 h within a particular medium ($p \geq 0.05$).

sions and estimated a half-life of < 5 h for citrate, indicating its fast degradation by microbes. Jones and Darrah (1992^[20]) reported selective re-absorption (retrieval) of specific components exuded by the roots after prolonged exposure, resulting in a loss and an alteration of the exudate spectrum. Jones and Darrah (1993^[21]) suggested the possibility of simultaneous exudation and re-absorption of soluble C compounds by maize roots. However, as the trends in the amounts of organic acids and carbohydrates in exudates collected after 2, 4 or 6 h were similar in the present experiments (Fig. 2), selective absorp-

tion of specific components exuded do not appear to be responsible for reduced exudate contents observed with prolonged exposure. The experimental set up of the present study was not designed to pinpoint the causal mechanism for the decrease in exudation with increasing collection time. However, our results clearly indicate that short-term collection is essential for minimizing the effects of the above-mentioned factors to avoid underestimation of C exuded by rice roots.

In summary, CaSO_4 solution may be regarded a preferential medium for collecting root exudates from rice or other plants. This medium maintains the osmotic environment of the root cells and – at the same time – does not interfere in analytical procedures, such as the wet digestion method for TOC and HPLC for organic acids. Furthermore, the collection of root exudates for a short period of time, such as 2 h, was found to be the best option.

Influence of plant growth and cultivars on root exudation

The root exudates of IR72 cultivar collected in 0.01 M CaSO_4 for 2 h revealed that TOC released by 31-d-old rice plants at seedling stage was $2.32 \text{ mg C plant}^{-1} \text{ d}^{-1}$ and increased by about 6-fold at maximum tillering stage (Fig. 2).

The TOC released by roots of tall, dwarf, NPT and hybrid plants are presented in Fig. 3 and Tables 3a,b. In general, the exudation rates were lowest at seedling stage (plant age 25–30 d) with mean values ranging between $0.6 \text{ mg C plant}^{-1} \text{ d}^{-1}$ (IR52) and $2.7 \text{ mg C plant}^{-1} \text{ d}^{-1}$ (Dular). At PI (plant age 60–70 d), release of TOC increased by a factor of 2 to 6 in different cultivars. Measurements at flowering revealed exudation rates ranging from 3.8 to $8.1 \text{ mg C plant}^{-1} \text{ d}^{-1}$. Among tall cultivars, Intan showed the highest release of TOC, followed by Dular and B40. High-yielding cultivars showed exudation in the descending order of IR64 > IR52 > IR72 > PSBRc 20. At flowering, dwarf cultivars (IR72, IR52, IR64 and PSBRc 20) exhibited a greater potential for root exudation (5.8 to $8.1 \text{ mg C plant}^{-1} \text{ d}^{-1}$) than traditional tall cultivars (5.6 to $6.0 \text{ mg C plant}^{-1} \text{ d}^{-1}$) and NPT cultivars (3.8 to $6.2 \text{ mg C plant}^{-1} \text{ d}^{-1}$). Both NPT cultivars (IR65598 and IR65600) showed very low TOC at the four growth stages studied. Hybrid Magat showed highest release of TOC at PI ($6.6 \text{ mg C plant}^{-1} \text{ d}^{-1}$) with no further change at flowering and maturity. On a unit biomass basis, large varia-

Table 2 Root and shoot (aboveground) biomass and amount of organic acids in root exudates of rice plants (IR72 cultivar) collected in CaSO_4 solution or deionized water for 2 h at seedling and maximum tillering stages

Collection media	Root biomass [g plant ⁻¹]	Shoot biomass	Malic acid [$\mu\text{mol plant}^{-1} \text{ d}^{-1}$]	Tartaric acid	Succinic acid	Citric acid
Seedling						
CaSO_4 solution	0.2 ± 0.0^a	1.0 ± 0.1	14.0 ± 0.7	0.8 ± 0.2	nd	0.2 ± 0.1
Deionized H_2O	0.2 ± 0.0	0.8 ± 0.0	13.2 ± 1.2	0.6 ± 0.1	0.9 ± 0.0	nd
LSD (0.05)	ns	ns	ns	ns	–	–
Maximum tillering						
CaSO_4 solution	2.6 ± 0.4	11.0 ± 0.4	135.9 ± 18.6	6.9 ± 2.1	9.4 ± 4.1	nd
Deionized H_2O	2.6 ± 0.3	11.3 ± 0.3	30.3 ± 4.3	2.7 ± 1.2	1.4 ± 0.2	nd
LSD (0.05)	ns	ns	6.8	1.5	2.0	–

^a Standard deviation; nd = not detected; ns = not significant.

Tables 3 a, b Organic acids in root exudates of rice plants of traditional tall cultivars (Dular, B40 and Intan), high-yielding dwarf cultivars (IR72, IR52 and IR64 and PSBRc 20), new plant type cultivars (IR65598 and IR65600) and a hybrid (Magat) at four growth stages. Note different units used for individual organic acids and total organic acids

a Cultivar	Malic acid				Tartaric acid			
	Seedling	Panicle initiation	Flowering	Maturity	Seedling	Panicle initiation	Flowering	Maturity
	[$\mu\text{mol plant}^{-1} \text{d}^{-1}$]							
Dular	11.1 $\pm$ 1.5 ^a	45.6 $\pm$ 4.5	53.1 $\pm$ 2.6	50.0 $\pm$ 2.9	3.7 $\pm$ 0.3	4.2 $\pm$ 0.6	5.0 $\pm$ 0.7	4.6 $\pm$ 0.3
Intan	6.1 $\pm$ 0.5	51.2 $\pm$ 4.6	65.1 $\pm$ 7.1	45.9 $\pm$ 1.0	2.4 $\pm$ 0.3	6.7 $\pm$ 0.2	6.4 $\pm$ 0.8	7.3 $\pm$ 0.4
B40	6.9 $\pm$ 0.5	52.5 $\pm$ 2.0	63.4 $\pm$ 7.9	43.5 $\pm$ 3.9	1.3 $\pm$ 0.2	6.1 $\pm$ 0.9	5.0 $\pm$ 0.5	7.3 $\pm$ 0.5
IR72	13.9 $\pm$ 0.5	63.9 $\pm$ 7.1	69.8 $\pm$ 14.6	30.7 $\pm$ 2.3	1.8 $\pm$ 0.1	6.4 $\pm$ 1.2	7.2 $\pm$ 0.3	6.8 $\pm$ 0.5
IR52	3.3 $\pm$ 0.5	27.4 $\pm$ 5.0	64.9 $\pm$ 2.3	65.4 $\pm$ 5.6	0.3 $\pm$ 0.0	1.3 $\pm$ 0.2	6.6 $\pm$ 0.4	6.7 $\pm$ 0.4
IR64	3.8 $\pm$ 0.2	37.2 $\pm$ 8.0	78.4 $\pm$ 10.6	58.2 $\pm$ 3.8	1.3 $\pm$ 0.2	8.8 $\pm$ 1.0	6.2 $\pm$ 0.7	6.0 $\pm$ 0.8
PSBRc 20	6.2 $\pm$ 1.9	43.8 $\pm$ 13.6	74.3 $\pm$ 11.1	61.0 $\pm$ 8.8	0.5 $\pm$ 0.1	3.5 $\pm$ 0.7	6.4 $\pm$ 0.6	6.8 $\pm$ 0.1
IR65598	6.8 $\pm$ 0.7	35.8 $\pm$ 0.2	47.6 $\pm$ 2.4	31.5 $\pm$ 1.9	1.1 $\pm$ 0.5	3.6 $\pm$ 0.3	3.7 $\pm$ 0.2	1.7 $\pm$ 0.1
IR65600	7.9 $\pm$ 1.6	23.9 $\pm$ 7.9	36.5 $\pm$ 5.0	38.9 $\pm$ 4.9	1.1 $\pm$ 0.7	2.6 $\pm$ 0.3	7.3 $\pm$ 0.7	6.0 $\pm$ 0.9
Magat	5.8 $\pm$ 2.4	43.8 $\pm$ 7.0	61.2 $\pm$ 2.9	61.8 $\pm$ 7.1	0.9 $\pm$ 0.0	7.1 $\pm$ 1.4	7.5 $\pm$ 0.7	6.8 $\pm$ 0.5
LSD (0.05)	2.4	4.0	4.1	4.7	0.7	1.4	2.1	2.3

^a Standard deviation

b Cultivar	Succinic acid				Citric acid		Lactic acid	Total organic acids			
	Seedling	Panicle initiation	Flowering	Maturity	Seedling	Panicle initiation	Panicle initiation	Seedling	Panicle initiation	Flowering	Maturity
	[$\mu\text{mol plant}^{-1} \text{d}^{-1}$]							[mg C plant ⁻¹ d ⁻¹]			
Dular	4.0 $\pm$ 0.8 ^a	4.8 $\pm$ 0.7	6.3 $\pm$ 0.4	4.3 $\pm$ 0.1	nd	1.9 $\pm$ 0.2	2.4 $\pm$ 0.4	0.9 $\pm$ 0.1	2.8 $\pm$ 0.2	3.1 $\pm$ 0.2	2.8 $\pm$ 0.1
Intan	2.0 $\pm$ 0.2	12.5 $\pm$ 0.5	12.9 $\pm$ 0.5	6.9 $\pm$ 0.3	1.6 $\pm$ 0.5	0.0 $\pm$ 0.0	1.2 $\pm$ 0.7	0.6 $\pm$ 0.0	3.4 $\pm$ 0.2	4.1 $\pm$ 0.3	2.9 $\pm$ 0.1
B40	2.1 $\pm$ 0.1	3.9 $\pm$ 0.7	5.0 $\pm$ 0.1	6.4 $\pm$ 0.3	0.0 $\pm$ 0.0	2.6 $\pm$ 0.3	2.8 $\pm$ 1.0	0.5 $\pm$ 0.0	3.3 $\pm$ 0.1	3.5 $\pm$ 0.4	2.7 $\pm$ 0.2
IR72	1.2 $\pm$ 0.1	nd	4.8 $\pm$ 0.4	6.6 $\pm$ 0.5	0.2 $\pm$ 0.0	9.2 $\pm$ 0.8	nd	0.8 $\pm$ 0.0	4.0 $\pm$ 1.0	3.9 $\pm$ 0.7	2.1 $\pm$ 0.1
IR52	0.6 $\pm$ 0.0	nd	9.8 $\pm$ 0.3	7.1 $\pm$ 0.4	nd	2.5 $\pm$ 0.1	nd	0.2 $\pm$ 0.0	1.6 $\pm$ 0.2	3.9 $\pm$ 0.1	3.8 $\pm$ 0.3
IR64	1.6 $\pm$ 0.4	nd	5.3 $\pm$ 0.5	3.7 $\pm$ 0.2	1.7 $\pm$ 0.7	2.4 $\pm$ 0.1	nd	0.4 $\pm$ 0.1	2.4 $\pm$ 0.3	4.3 $\pm$ 0.5	3.3 $\pm$ 0.2
PSBRc 20	1.1 $\pm$ 0.1	nd	4.3 $\pm$ 0.1	3.6 $\pm$ 0.1	0.8 $\pm$ 0.2	4.9 $\pm$ 0.1	nd	0.4 $\pm$ 0.1	2.6 $\pm$ 0.6	4.1 $\pm$ 0.5	3.4 $\pm$ 0.4
IR65598	3.3 $\pm$ 0.9	3.2 $\pm$ 0.6	4.9 $\pm$ 0.3	3.6 $\pm$ 0.0	nd	nd	nd	0.5 $\pm$ 0.0	2.1 $\pm$ 0.0	2.7 $\pm$ 0.1	1.7 $\pm$ 0.1
IR65600	2.4 $\pm$ 0.2	2.7 $\pm$ 0.1	9.3 $\pm$ 0.8	3.3 $\pm$ 0.2	nd	nd	nd	0.6 $\pm$ 0.1	1.4 $\pm$ 0.4	2.6 $\pm$ 0.2	2.3 $\pm$ 0.2
Magat	nd	nd	10.9 $\pm$ 1.3	10.3 $\pm$ 0.1	2.6 $\pm$ 0.1	9.7 $\pm$ 0.2	nd	0.5 $\pm$ 0.1	3.2 $\pm$ 0.3	3.8 $\pm$ 0.1	3.8 $\pm$ 0.9
LSD (0.05)	0.6	1.1	2.3	2.0	1.0	2.1	0.8	0.4	2.2	1.3	1.5

^a Standard deviation; nd = not detected.

tions in exudation rates, ranging from 0.36 to 5.67 mg g⁻¹ root biomass d⁻¹ and 0.10 to 1.06 mg g⁻¹ root + shoot biomass d⁻¹, were observed among growth stages. However, there was no consistent trend of differences at different growth stages.

These results indicate that root exudation of all cultivars increased with plant growth until PI but differed in the succeeding growth stages. While 4 of the 10 cultivars showed significant increase at flowering (IR52, IR64, PSBRc 20 and IR65600), the remaining cultivars exhibited no significant change. A decreasing trend in exudation from flowering to maturity was evident in all of the 9 inbred cultivars, though the decrease was significant in only 5 cultivars (Dular, Intan, B40, IR72 and IR65598). Sauerbeck and Johnen (1976^[38]) reported an increase in release of organic C by the roots of soil-grown wheat resulting in highest exudation during the period of ra-

pid vegetative growth. Differences in both exudate quality and quantity with plant age have previously been reported for other plant species (Vanura et al., 1977^[40]; Hale and Moore, 1979^[15]; Pikryl and Vanura, 1980^[35]). The decrease in root excretion at maturity of inbred cultivars observed in the present study may be due to death of some root cells (Aulakh et al., 2000^[41]).

Influence of plant growth and cultivars on the composition of root exudates

Individual organic acids present in the exudates showed significant differences in quantities and trends among cultivars, as well as during plant growth. Malic acid was present in conspicuously large amounts (3.3 to 78.4 $\mu\text{mol plant}^{-1} \text{d}^{-1}$), showing an increasing trend with plant growth from seedling to

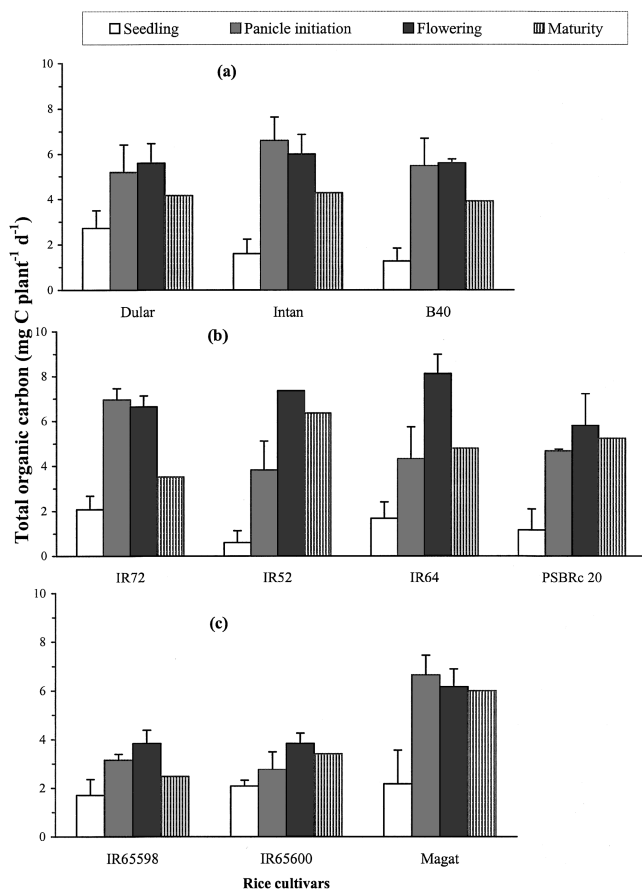


Fig. 3 Root exudation (total organic C) of rice plants of (a) traditional tall cultivars (Dular, B40 and Intan), (b) high-yielding dwarf cultivars (IR72, IR52, IR64 and PSBRc 20), (c) new plant type cultivars (IR65598 and IR65600) and a hybrid (Magat) at four growth stages. Data shown are means $\pm$ SD of 3 replicate plants.

flowering in all cultivars (Tables 3a,b). At maturity, its amount did not change significantly in 4 cultivars (Dular, IR52, IR65600 and Magat) and decreased significantly in the remaining 6 cultivars. Tartaric acid was 3 to 19 times less in quantity than malic acid but exhibited similar increasing trends with plant growth (Tables 3a,b). At maturity, however, only IR65598 showed a decrease in the amount of tartaric acid exuded. Succinic acid followed similar trends but was not detected in Magat at seedling and PI, or in IR72, IR52, IR64 and PSBRc 20 at PI (Tables 3a,b). The level of succinic acid decreased significantly at maturity in 5 cultivars (Dular, Intan, IR52, IR64 and IR65600). Citric acid was present in only 5 cultivars at seedling, 8 at PI and none at flowering and maturity (Tables 3a,b). Moreover, its amount was 2 to 70 times lower than malic, 2 to 4 times greater than tartaric and 2 to 6 times greater than succinic acid. Lactic acid was detected only at PI in 3 tall cultivars with its amount ranging from 1.2 to 2.8 $\mu\text{mol plant}^{-1} \text{d}^{-1}$. From seedling to maximum tillering stage of IR72, the amount of malic and tartaric acids increased by 10- and 8-fold, respectively; 0.23 $\mu\text{mol plant}^{-1} \text{d}^{-1}$ of citric acid was exuded at seedling stage but none at maximum tillering (Tables 3a,b). Malic acid in different cultivars from seedling to flowering increased between 5- to 21-fold; for tartaric acid, the corresponding increase ranged from 1- to 22-fold.

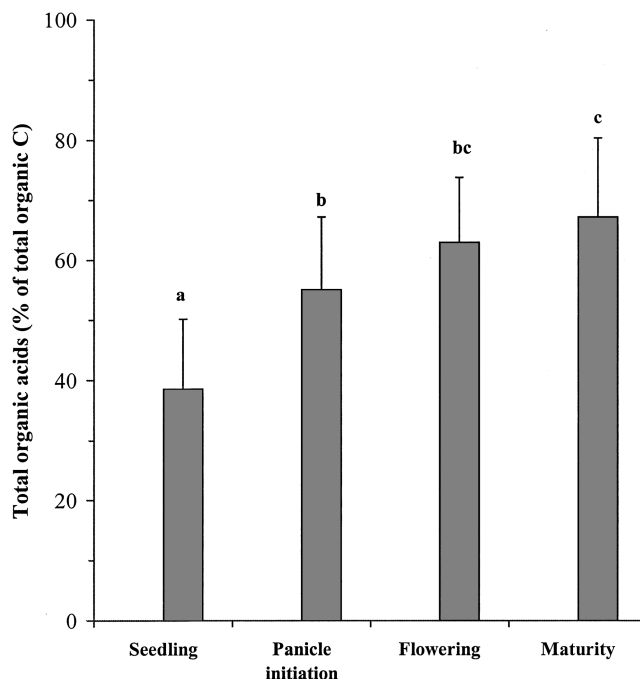


Fig. 4 Proportion of organic acids in total organic C released from the roots of rice plants at four growth stages (average of 10 cultivars). Data shown are means $\pm$ SD of 30 plants (10 cultivars $\times$ 3 replicates). Different letters indicate significant differences between growth stages ($p \geq 0.05$).

The total amount of organic acids showed a 3- to 20-fold increase from seedling to flowering with either no further change or decrease at maturity (Fig. 4, Table 4). The proportion of organic acids in TOC also continued to increase with plant growth in all cultivars. On average in 10 cultivars, its proportion was 39% at seedling and increased to 67% at maturity (Fig. 4). Lin and You (1989^[26]) examined the root exudates of hydroponically-grown rice plants and reported a total of 2.25 mg organic acids and 1.27 mg carbohydrates g^{-1} fresh root. In our experiment with the IR72 cultivar at seedling and maximum tillering, the proportion of organic acids increased from 31 to 54% with a simultaneous decrease in the proportion of carbohydrates from 65 to 43% of TOC (Fig. 2), suggesting the substitution of carbohydrates by organic acids. The results covering the period from seedling to maturity further confirmed the increase in the proportion of organic acids in root exudates as plant growth advances. However, more studies are needed to confirm the metabolic interconversion of the carbohydrates to organic acids. The unaccounted portion of about 2 to 4% of TOC presumably represented other organic compounds, such as amino acids, often present in root exudates (Mac Rae and Castro, 1966^[30]; Marschner, 1995^[31]) but not analysed in this study.

These results showed (a) that significant differences in the amount of TOC and individual organic acids exuded exist between individual rice cultivars at a given growth stage; (b) that among the organic acids, malic acid has the highest concentration in rice root exudates followed by tartaric, succinic, citric and lactic acids in decreasing order; and (c) that exudation rates, in general, increase with plant development from seed-

Table 4 Results of analysis of variance and mean exudation of total organic carbon; different letters indicate significant differences (0.05 level of probability) across growth stages (for given cultivar) and across cultivars (at given growth stage); S = seedling, P = panicle initiation, F = flowering, M = maturity. Different letters indicate significant differences at $p \geq 0.05$

Cultivar	Across growth stages				Across cultivars				Mean exudation [mg C plant ⁻¹ d ⁻¹]
	S	P	F	M	S	P	F	M	
Dular	a	bc	c	b	c	c	b	bc	4.4
Intan	a	c	c	b	b	d	b	bc	4.6
B40	a	c	c	b	a	cd	b	b	4.07
IR72	a	c	c	b	bc	d	bc	b	4.8
IR52	a	b	c	c	a	abc	cd	d	4.54
IR64	a	b	c	c	a	bc	d	cd	4.73
PSBRc 20	a	b	c	bc	a	c	b	cd	4.22
IR65598	a	bc	c	ab	b	ab	a	a	2.8
IR65600	a	a	b	ab	bc	a	a	ab	3.02
Magat	a	b	b	b	bc	d	bc	d	5.24

Table 5 Root and shoot (aboveground) biomass at four growth stages and grain yield of traditional tall cultivars (Dular, B40 and Intan), high-yielding dwarf cultivars (IR72, IR52 and IR64 and PSBRc 20), new plant type cultivars (IR65598 and IR65600) and a hybrid Magat

Cultivar	Root biomass				Shoot biomass				Grain yield at maturity
	Seedling [g plant ⁻¹]	Panicle initiation	Flowering	Maturity	Seedling	Panicle initiation	Flowering	Maturity	
Dular	0.5 ± 0.1 ^a	2.8 ± 0.3	4.8 ± 0.7	4.9 ± 0.8	2.1 ± 0.2	7.5 ± 0.4	21.4 ± 3.3	16.1 ± 3.2	11.7 ± 0.6
Intan	0.6 ± 0.2	5.3 ± 0.8	7.8 ± 1.5	9.0 ± 1.5	2.1 ± 0.2	27.4 ± 1.4	41.5 ± 2.0	33.8 ± 3.4	11.5 ± 0.9
B40	0.5 ± 0.1	2.8 ± 0.3	3.8 ± 0.7	4.1 ± 0.1	1.7 ± 0.2	8.6 ± 1.0	20.0 ± 3.3	17.3 ± 0.6	9.4 ± 1.0
IR72	0.6 ± 0.1	5.2 ± 0.1	4.1 ± 1.1	3.9 ± 0.8	1.9 ± 0.1	13.6 ± 1.7	22.0 ± 2.1	14.2 ± 0.7	15.8 ± 1.3
IR52	0.6 ± 0.2	5.5 ± 1.2	4.1 ± 0.8	4.9 ± 0.2	1.7 ± 0.2	14.7 ± 2.5	21.6 ± 1.7	17.6 ± 1.4	15.9 ± 0.8
IR64	0.6 ± 0.2	3.3 ± 0.4	4.6 ± 0.4	4.1 ± 0.2	1.7 ± 0.4	12.8 ± 2.0	23.2 ± 1.1	19.1 ± 2.1	16.9 ± 2.4
PSBRc 20	0.6 ± 0.2	3.9 ± 0.6	3.2 ± 0.4	3.1 ± 0.6	1.8 ± 0.3	11.8 ± 0.3	16.5 ± 2.6	13.7 ± 0.7	13.8 ± 1.0
IR65598	0.5 ± 0.2	4.1 ± 0.6	4.5 ± 0.2	3.7 ± 0.6	1.9 ± 0.3	16.0 ± 1.7	21.2 ± 1.8	18.2 ± 1.3	10.3 ± 1.8
IR65600	0.3 ± 0.1	4.8 ± 0.3	5.0 ± 0.2	4.8 ± 0.3	1.6 ± 0.2	19.6 ± 1.2	23.2 ± 2.6	21.8 ± 2.1	10.6 ± 0.7
Magat	0.7 ± 0.2	4.9 ± 0.5	4.1 ± 0.4	3.4 ± 0.5	2.6 ± 0.2	14.3 ± 1.6	20.5 ± 2.0	14.5 ± 3.2	14.3 ± 3.3
LSD (0.05)	ns	1.0	1.8	1.5	ns	3.2	3.0	4.3	2.6

^a Standard deviation; ns = not significant.

ling to PI or flowering but decrease at maturity, except for the hybrid Magat.

Relationships between root exudation and plant parameters

The growth parameters of the 10 cultivars tested (Table 5) revealed large differences among cultivars. The root biomass of these cultivars did not vary significantly at seedling stage but was largest in Intan and lowest in PSBRc 20 at flowering and maturity. Similar trends were observed for shoot biomass. Plants continued to accumulate both root and shoot biomass with plant growth until flowering. At maturity, there was a substantial increase in root biomass of Intan, a decrease in IR64, IR65598 and Magat, and no change in the remaining 6 cultivars. However, shoot biomass, in general, declined at maturity. Grain yields of dwarf cultivars (IR72, IR52, IR64 and PSBRc 20) were significantly higher than for the rest of the cultivars (Table 5).

Root exudation during different growth stages showed a highly significant positive relationship with root biomass in all cultivars with the r values of individual cultivars ranging from 0.619 to 0.888 ($p \geq 0.01$; $n = 12$). The exudation rate also exhibited a strong relationship with shoot biomass ($r = 0.593$ to 0.929) and the root + shoot biomass ($r = 0.605$ to 0.930). These relationships obtained with individual cultivars were also found with the pooled analysis of 10 cultivars (Fig. 5). From the improvement in the correlation, it appeared that the amount of exudate C released by the roots is associated with both root and aboveground plant biomass. We found a highly significant correlation of the root biomass with the shoot biomass ($r = 0.906^{**}$, $n = 120$). These results suggest that high aboveground dry matter production is associated with a large root system resulting in increased TOC releases from the roots.

The decrease in the correlation coefficient from 0.626 with the root + shoot biomass to 0.553 with the root + shoot + grain biomass (Fig. 5) suggests that grain yield has a negative effect on

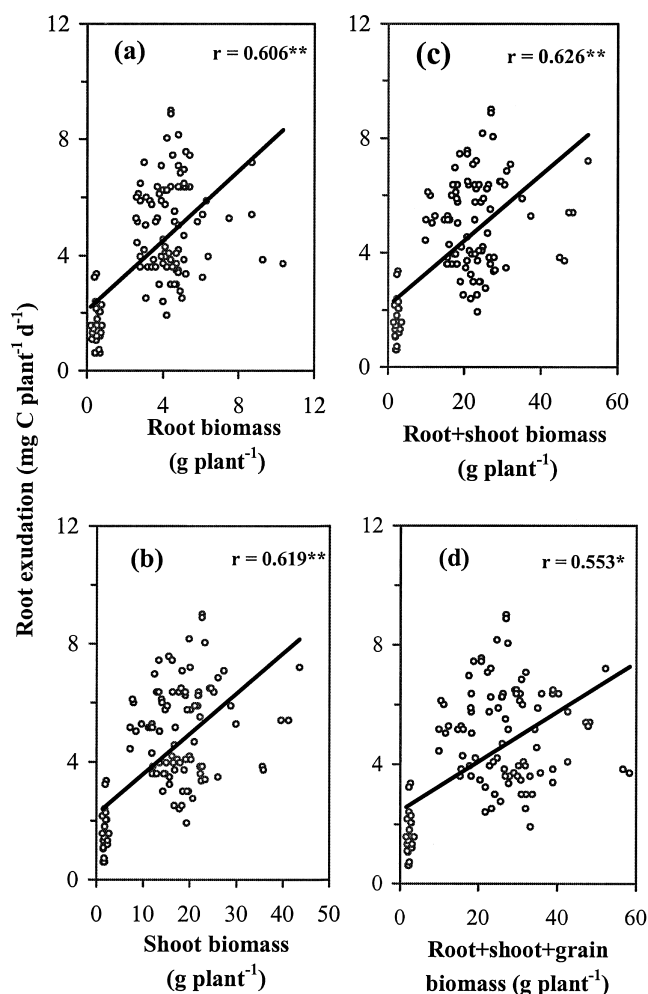


Fig. 5 Relationship of root exudation rate to (a) root biomass, (b) shoot (aboveground) biomass, (c) root + shoot biomass and (d) root + shoot + grain biomass of rice plants. Pooled data ($n = 120$) for 10 cultivars and 4 growth stages.

root exudation. This is desirable because minimizing the loss of photosynthetic C from the plant via root exudation would result in increased rice yields. For example, NPT cultivars that are developed to produce high yields showed lowest C exudation.

Conclusions

The use of CaSO_4 solution as a medium for collecting root exudates of rice plants grown under natural flooded soil conditions was found to be a useful tool for obtaining information on the quantity and composition of root exudates of different cultivars at different growth stages. This method could be employed in future studies to assess the role of exudate-induced changes in the rhizosphere, in order to understand the mechanisms by which rice plants adapt to nutrient-deficient situations (e.g., P deficiency) or to adverse soil conditions, such as high Al concentrations. Since the root-exuded organic compounds act as substrates for methanogenesis (Dannenberg and Conrad, 1999^[11]; Lu et al., 2000^[29]), and there is a large variation among the methane transport capacity among rice cul-

tivars (Aulakh et al., 2000a^[2], 2000b^[3]), future studies on the impact of root exudates on the production of CH_4 from rice agriculture are highly desirable.

Acknowledgements

We thank Sonny Pantoja for help with the greenhouse experiments. This research was funded by the German BMZ/GTZ within the frame of the project "Reduction of methane emission from rice fields by screening for low methane transport capacity" (No. 95.7860.0-001.05).

References

- Andal, R., Bhuvanesware, K., and Subba-Rao, N. (1956) Root exudates of paddy. *Nature* 178, 1063.
- Aulakh, M. S., Bodenbender, J., Wassmann, R., and Rennenberg, H. (2000a) Methane transport capacity of rice plants. I. Influence of methane concentration and growth stage analyzed with an automated measuring system. *Nut. Cycling Agroecosyst.* 58, 357–366.
- Aulakh, M. S., Bodenbender, J., Wassmann, R., and Rennenberg, H. (2000b) Methane transport capacity of rice plants. II. Variations among different rice cultivars and relationship with morphological characteristics. *Nut. Cycling Agroecosyst.* 58, 367–375.
- Aulakh, M. S., Wassmann, R., Rennenberg, H., and Fink, S. (2000c) Pattern and amount of aerenchyma relates to variable methane transport capacity of different rice cultivars. *Plant Biol.* 2, 182–194.
- Aulakh, M. S., Wassmann, R., Bueno, C., and Rennenberg, H. (2001) Impact of root exudates of different rice cultivars on methane production in soil under anaerobic conditions. *Plant and Soil*, in press.
- Badoud, R. and Pratz, G. (1986) Improved high-performance liquid chromatographic analysis of some carboxylic acids in food and beverages as their p-nitrobenzyl esters. *J. Chromatography* 360, 119–136.
- Bartolome, V. I., Casumpang, R. M., Ynalvez, M. A. H., Olea, A. B., and McLaren, C. G. (1999) IRRISTAT for Windows – Statistical Software for Agricultural Research. Los Banos, The Philippines: Biometrics, International Rice Research Institute.
- Beissner, L. and Römer, W. (1998) Mobilization of phosphorus by root exudates of sugar beet. In *Summaries, 16th World Congress Soil Science, Vol. II. Montpellier, France*, p. 776.
- Brink, R. H. Jr., Dubach, P., and Lynch, D. L. (1960) Measurement of carbohydrates in soil hydrolyzates with anthrone. *Soil Sci.* 89, 157–166.
- Cochran, W. G. and Cox, G. M. (1950) *Experimental Designs*. New York: John Wiley and Sons, Inc.
- Dannenberg, S. and Conrad, R. (1999) Effect of rice plants on methane production and rhizospheric metabolism in paddy soil. *Bio-geochem.* 45, 53–71.
- Dinkelaker, B., Hengeler, C., Neumann, G., and Marschner, H. (1996) Root exudates and mobilization of nutrients. In *Trees – Contributions to Modern Tree Physiology* (Rennenberg, H., Eschrich, W., and Ziegler, H., eds.), Amsterdam: Academic Publishing bv, pp. 3–14.
- Fukumori, T. and Chino, M. (1982) Sugar, amino acid and inorganic contents in rice phloem sap. *Plant Cell Physiol.* 23, 273–283.
- Giaquinta, R. (1980) Mechanism and control of phloem loading of sucrose. *Ber. Deutsch Bot. Ges.* 93, 187–201.
- Hale, M. G. and Moore, L. D. (1979) Factors affecting root exudation. II: 1970–1978. *Adv. Agron.* 31, 93–124.
- Hodge, A., Grayston, S. J., and Ord, B. G. (1996) A novel method for characterisation and quantification of plant root exudates. *Plant and Soil* 184, 97–104.

- ¹⁷ Hoffland, E. E., van den Boogaard, R., Nelemans, J. A., and Findenegg, G. R. (1989) Solubilization of rock phosphate by rape. II. Local root exudation of organic acids as a response to P starvation. *Plant and Soil* 113, 161–165.
- ¹⁸ Holzapfel-Pschorn, A., Conrad, R., and Seiler, W. (1986) Effects of vegetation on the emission of methane from submerged paddy soil. *Plant and Soil* 92, 223–391.
- ¹⁹ Jauregui, M. A. and Reisenauer, H. M. (1982) Dissolution of oxides of manganese and iron by root exudate components. *Soil Sci. Soc. Am. J.* 46, 314–317.
- ²⁰ Jones, D. L. and Darrah, P. R. (1992) Re-sorption of organic compounds by roots of *Zea mays* L. and its consequences in the rhizosphere. I. Re-sorption of ¹⁴C labelled glucose, mannose and citric acid. *Plant and Soil* 143, 259–266.
- ²¹ Jones, D. L. and Darrah, P. R. (1993) Re-sorption of organic compounds by roots of *Zea mays* L. and its consequences in the rhizosphere. II. Experimental and model evidence for simultaneous exudation and re-sorption of soluble C compounds. *Plant and Soil* 153, 47–59.
- ²² Jones, D. L. and Darrah, P. R. (1994) Role of root derived organic acids in the mobilization of nutrients from the rhizosphere. *Plant and Soil* 166, 247–257.
- ²³ Kirk, G. J. D., Santos, E. E., and Santos, M. B. (1999) Phosphate solubilization by organic anion excretion from rice growing in aerobic soil: rates of excretion and decomposition, effects on rhizosphere and pH and effects on phosphate solubility and uptake. *New Phytol.* 142, 185–200.
- ²⁴ Kludze, H. K., DeLaune, R. D., and Patrick, W. H. Jr. (1993) Aerenchyma formation and methane and oxygen exchange in rice. *Soil Sci. Soc. Am. J.* 57, 386–391.
- ²⁵ Leon, J. C. and Carpena, A. L. (1995) Morphology-based diversity analysis of improved irrigated lowland rice (*Oryza sativa* L.) varieties in the Philippines. *Philipp. J. Crop Sci.* 20 (2), 113–121.
- ²⁶ Lin, M. and You, C. (1989) Root exudates of rice (*Oryza sativa* L.) and its interaction with *Alcaligenes faecalis*. *Scientia Agricultura Sinica* 22, 6–12 (in Chinese).
- ²⁷ Lindau, C. W., Bollich, P. K., DeLaune, R. D., Patrick, W. H. Jr., and Law, V. J. (1991) Effect of urea fertilizer and environmental factors on CH₄ emissions from a Louisiana USA rice field. *Plant and Soil* 136, 195–203.
- ²⁸ Lu, Y., Wassmann, R., Neue, H. U., and Huang, C. (1999) Impact of phosphorus supply on root exudation, aerenchyma formation and methane emission of rice plants. *Biogeochem.* 47, 203–218.
- ²⁹ Lu, Y., Wassmann, R., Neue, H. U., Huang, C., and Bueno, C. S. (2000) Methanogenic responses to exogenous substrates in anaerobic rice soils. *Soil Biol. Biochem.* 32, 1683–1690.
- ³⁰ Mac Rae, I. C. and Castro, T. F. (1966) Carbohydrates and amino acids in the root exudates of rice seedlings. *Phyton* 23, 95–100.
- ³¹ Marschner, H. (1995) *Mineral Nutrition of Higher Plants*, 2nd Edition. London: Academic Press.
- ³² Marschner, H., Treeby, M., and Kissel, M. (1989) Role of root-induced changes in the rhizosphere for iron acquisition in higher plants. *Z. Pflanzenernähr. Bodenk.* 152, 197–204.
- ³³ Nelson, D. W. and Sommers, L. E. (1996) Total carbon, organic carbon, and organic matter. In *Methods of Soil Analysis, Part 3 – Chemical methods*. SSSA Book Series no. 5 (Sparks, D. L., ed.), Madison, Wisconsin: Soil Science Society of America and American Society of Agronomy, pp. 961–1010.
- ³⁴ Neumann, G. and Römhild, V. (1998) Phosphorus deficiency-induced root excretion of carboxylic acids and protons in different plant species. In *Summaries, 16th World Congress Soil Science, Vol. II*. Montpellier, France, p. 783.
- ³⁵ Pikryl, Z. and Vanura, V. (1980) Root exudates of plants. VI. Wheat root exudation as dependent on growth, concentration gradient of exudates and the presence of bacteria. *Plant and Soil* 57, 69–83.
- ³⁶ Rattray, E. A. S., Paterson, E., and Killham, K. (1995) Characterisation of the dynamics of C-partitioning within *Lolium perenne* and to the rhizosphere microbial biomass using ¹⁴C pulse chase. *Biol. Fertil. Soils* 19, 280–286.
- ³⁷ Rennenberg, H., Wassmann, R., Papen, H., and Seiler, W. (1992) Trace gas emission in rice cultivation. *Ecol. Bull.* 42, 164–173.
- ³⁸ Sauerbeck, D. and Johnen, B. (1976) Der Umsatz von Pflanzenwurzeln im Laufe der Vegetationsperiode und dessen Beitrag zur "Bodenatmung". *Z. Pflanzenernähr. Bodenk.* 139, 315–328.
- ³⁹ Shen, Y., Ström, L., Jönsson, J. A., and Tyler, G. (1996) Low-molecular organic acids in the rhizosphere soil solution of beech forest (*Fagus sylvatica* L.) cambisols determined by ion chromatography using supported liquid membrane enrichment technique. *Soil Biol. Biochem.* 28, 1163–1169.
- ⁴⁰ Vanura, V., Pikryl, Z., Kalachova, L., and Wurst, M. (1977) Some quantitative aspects of root exudates. *Ecol. Bull.* 25, 381–386.
- ⁴¹ Vaughan, D., Cheshire, M. V., and Ord, B. G. (1994) Exudation of peroxide from roots of *Festuca rubra* and its effects on exuded phenolic acids. *Plant and Soil* 160, 153–155.
- ⁴² Wang, B., Neue, H. U., and Samonte, H. P. (1997) Effect of rice plant on seasonal methane emission patterns. *Acta Agronomica Sinica* 23, 271–279.
- ⁴³ Wassmann, R. and Aulakh, M. S. (2000) The role of rice plants in regulating mechanisms of methane emission. *Biol. Fertil. Soils* 31, 20–29.
- ⁴⁴ Westerman, R. L. (1990) *Soil Testing and Plant Analysis*, Third Edition. Madison, Wisconsin: Soil Science Society of America.
- ⁴⁵ Zhang, F., Römhild, V., and Marschner, H. (1989) Effect of zinc deficiency in wheat on the release of zinc and iron mobilizing root exudates. *Z. Pflanzenernähr. Bodenk.* 152, 205–210.

H. Rennenberg

Institute for Forest Botany and Tree Physiology
Albert-Ludwigs University
79085 Freiburg
Germany
E-mail: here@uni-freiburg.de

Section Editor: U. Lüttge